HENRY GLOVER

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EDUCATION

University of Southern California *Dec 2026*
Master of Science, Aerospace Engineering – GPA: 4.0
- Viterbi Scholar - Dean's List

University of Southern California *Dec 2025*
Bachelor of Science, Mechanical Engineering – GPA: 3.3

SKILLS

Technical Skills: Siemens NX, SolidWorks, Abaqus, ANSYS, NASGRO, Simulink, GD&T, Microsoft Project/Suite, 3D Printing
Interpersonal Skills: Personable, Collaborative Soft Skills, Effective Communicator, Team Leadership, Detail-Oriented, Adaptable
Programming Languages: MATLAB, C++, LabVIEW, Arduino Microcontroller

PROJECTS

MACH-0 – Deployable Hip-Protection Device *Aug 2024 - Dec 2024*
- Designed a spring-steel deployment mechanism with a solenoid actuated buckling rod release in Siemens NX to develop a reusable mechanical alternative to airbag and foam hip protectors for elderly fall protection; achieved a reduction in peak impact force, exceeding commercial foam performance benchmarks
- Prepared technical deliverables and written documentation including component drawings with appropriate GD&T, progress and assessment reports, and supporting hand-calculations to guide concept design, verifying a predicted compression force against experimental measurements
- Characterized impact response using three test setups: a drop tower, a hydraulic press, and a custom pendulum jig with an OptiTrack motion capture system polling at 1000 Hz, processing the data in MATLAB and running Abaqus FEA to confirm the design remained below yield prior to fabrication
- Administered Microsoft Project to build Gantt-chart schedules that coordinated a four-person team and generated a bill of materials for budget estimation and vendor sourcing, keeping the build on schedule
- Built and validated a functional proof-of-concept prototype, then presented the results to senior faculty and over 100 department members, securing approval for long-term project continuation

Formula SAE Electric, Ergonomics & Vehicle Dynamics Team *Feb 2024 - Feb 2026*
- Collaborated with a team to design steering linkages, pedal box, steering wheel, and driver-interface assemblies in Siemens NX; engineered the steering linkage for tension and compression loads per FSAE requirements, designed and machined an aluminum pedal box to meet minimum clearance requirements, and iterated designs around evolving packaging constraints
- Integrated the braking system across two independent hydraulic circuits and designed the pedal box to withstand the mandated 2000 N of pedal force, sized the system to lock all four wheels and maintain braking under a single-point failure, and implemented an analog brake over-travel switch to trigger the shutdown circuit in compliance with FSAE requirements
- Performed system-level suspension and vehicle dynamics design, sizing shock absorbers based on vehicle mass and load transfer to meet minimum wheel travel requirements; designed and machined in-house A-arms and wishbones, and developed uprights and hubs in Siemens NX with structural validation using Abaqus and ANSYS

Smart Box – IoT Smart-Mailbox Notification System *Feb 2023 - Mar 2023*
- Designed and built a two-module IoT mailbox system using Particle Argon microcontrollers, integrating automated mail detection and real-time OTA alerts to notify users of deliveries and eliminate unnecessary mailbox checks
- Integrated three sensors: ultrasonic, magnetic switch, and motion sensor to detect door openings and confirm mail presence with redundant validation, reducing false positives from any single sensor
- Programmed both modules in C++ using VS Code with a publish/subscribe architecture, streaming a 15-signal telemetry feed to an initial state dashboard and phone application for live status and historical delivery analytics
- Developed a desk-side notification module with an OLED display driven by a mailbox-state machine, giving visual and audio alerts as well as configurable controls

Aerospace Structures & FEA Analysis *Jan 2024 - Dec 2025*
- Completed 20+ structural analysis projects in Abaqus, ANSYS, and NASGRO based on problem statements specifying geometry, material properties, loading, and boundary conditions; solving for loads, stresses, strains, and deflections and verifying against analytical solutions
- Analyzed aerospace hardware and structures, including a satellite under launch loads, pressurized fuselage skin, aircraft wings, composite pressure vessels, honeycomb sandwich panels, bolted joints, and welds; performed NASGRO fracture and fatigue-life analysis, as well as buckling and modal analysis
- Performed root-cause analysis using Fishbone diagrams and Failure Modes and Effects Analysis on aerospace case studies, identifying failure modes and damage-tolerance drivers; applied standardized fracture control methods using Linear Elastic Fracture Mechanics to evaluate stress intensity factors and critical crack sizes
- Designed and optimized composite laminates by applying Classical Laminate Theory to construct ABD stiffness matrices and compute ply-level stresses, evaluated failure with Tsai-Wu and maximum-strain criteria, and performed GRG optimization in Excel to minimize laminate weight under strength constraints for sandwich panels and pressure-vessel shells

WORK EXPERIENCE

Distribution Manager, Operations *Vinyl Vista, Thousand Oaks, CA* *Jan 2023 - Present*
- Manage end-to-end customer order fulfillment, coordinating shipments, postal services, and returns while analyzing inventory levels and demand to identify replenishment needs and guide purchasing decisions